BEYOND MULTIPLE LINEAR REGRESSION · WEEK 2

Treatment effects across centers

Interactions and uncertainty

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METHODS · DESIGN · STATISTICS

Week 1 recap: comparing the models

THE REHABILITATION TRIAL

Simulated trial: 120 patients after knee replacement at six centers.

Within each center: 10 patients randomized to each treatment.

Standard care: weekly physiotherapy and home exercises.

Enhanced care: standard care plus a weekly supervised session.

Outcome: six-minute walk distance at 12 weeks.

THE TREATMENT COMPARISON

Model	Estimate (m)	Estimated SE (m)
Treatment only	11.0	5.77
Fixed center effects	11.0	4.34
Random center effects	11.0	4.34

All three models assumed a common treatment effect across centers.

Fixed center parameters and a random center distribution gave the same treatment estimate and estimated standard error in this setting.

Changing treatment allocation within centers

ORIGINAL ALLOCATION

Center	Standard care	Enhanced care
A	10	10
B	10	10
C	10	10
D	10	10
E	10	10
F	10	10
Total	60	60

UNEQUAL ALLOCATION

Center	Standard care	Enhanced care
A	5	15
B	8	12
C	10	10
D	10	10
E	12	8
F	15	5
Total	60	60

Only allocation changes: the underlying treatment effect remains 10 m in every center.

The models now give different treatment estimates

Unequal allocation within centers; estimated enhanced-minus-standard difference in walk distance.

Model	Estimate (m)	Estimated SE (m)
Treatment only	3.87	5.75
Fixed center effects	8.40	4.56
Random center effects	8.13	4.55

Accounting for center changes the estimate substantially. The two center models give slightly different estimates, with almost identical estimated standard errors.

The models use different comparisons

TREATMENT ONLY

The model compares the average outcome of all enhanced-care patients with that of all standard-care patients.

Unequal allocation gives the groups different mixtures of centers. Their difference reflects both treatment and differences in center baselines.

FIXED CENTER EFFECTS

The model allows each center to have its own baseline outcome.

It estimates treatment by combining comparisons between enhanced and standard care within the same center.

These comparisons remove the shared center baseline.

RANDOM CENTER EFFECTS

The model uses comparisons both within and between centers.

Do centers with more enhanced care have higher average outcomes?

Using between-center information assumes baseline center effects are unrelated to treatment allocation.

The random-intercept model also uses between-center information under additional assumptions, so its treatment estimate can differ.

Fixed versus random center effects

Design	Treatment estimates	Estimated standard errors
Equal center sizes; 50:50 allocation within each center	Identical	Identical
Unequal center sizes; 50:50 allocation within each center	Identical	Can differ
Treatment proportions differ across centers	Can differ	Can differ

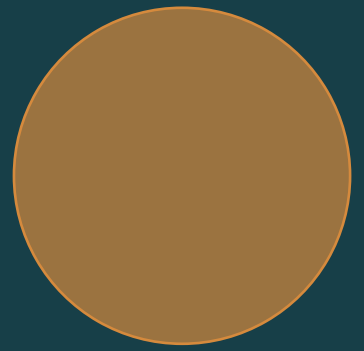
With 50:50 allocation within every center, all three models, including the treatment-only model, give the same treatment estimate, even when center sizes differ.

So far, we have assumed the same treatment effect in every center. What changes when the treatment effect varies across centers?

PART 1

Treatment-by-center interaction

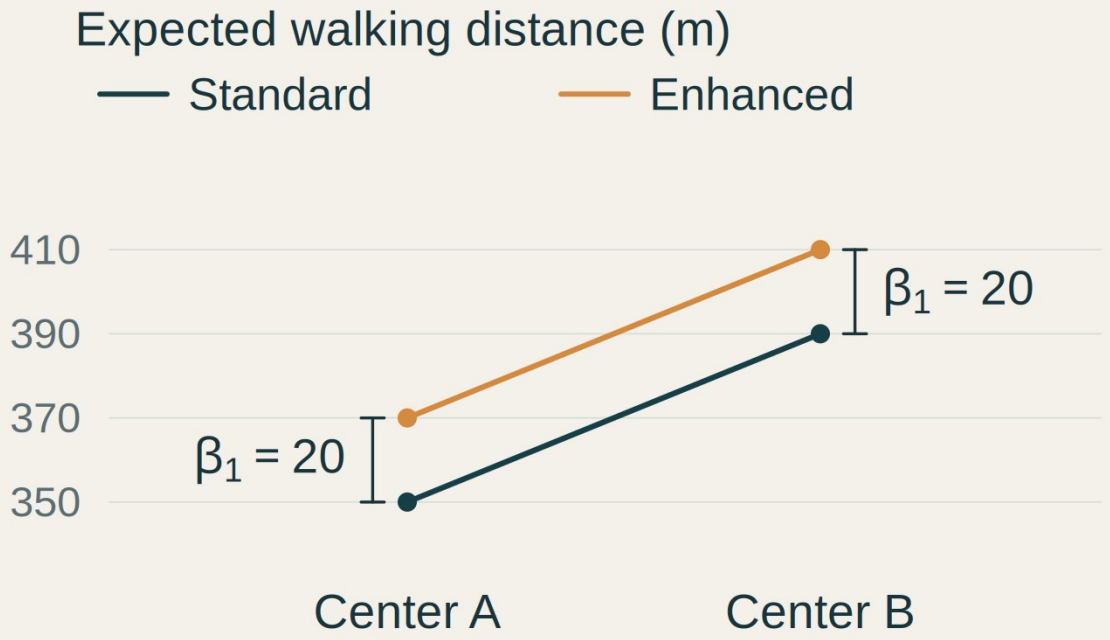




A common treatment effect in both centers

Hypothetical example: assume the four true group means are known.

Center	Standard	Enhanced	Row average
A	$\mu_{0A} = 350$	$\mu_{1A} = 370$	**360**
B	$\mu_{0B} = 390$	$\mu_{1B} = 410$	**400**
Column average	$\beta_0 = 370$	390	**380**



μ_{tj}: expected outcome for treatment t in center j

$$Y_{ij} = \underbrace{\beta_0 + c_j}_{\text{Standard-care mean in center } j} + \underbrace{\beta_1 T_{ij}}_{\text{Treatment contribution}} + \underbrace{\varepsilon_{ij}}_{\text{Individual deviation}}$$

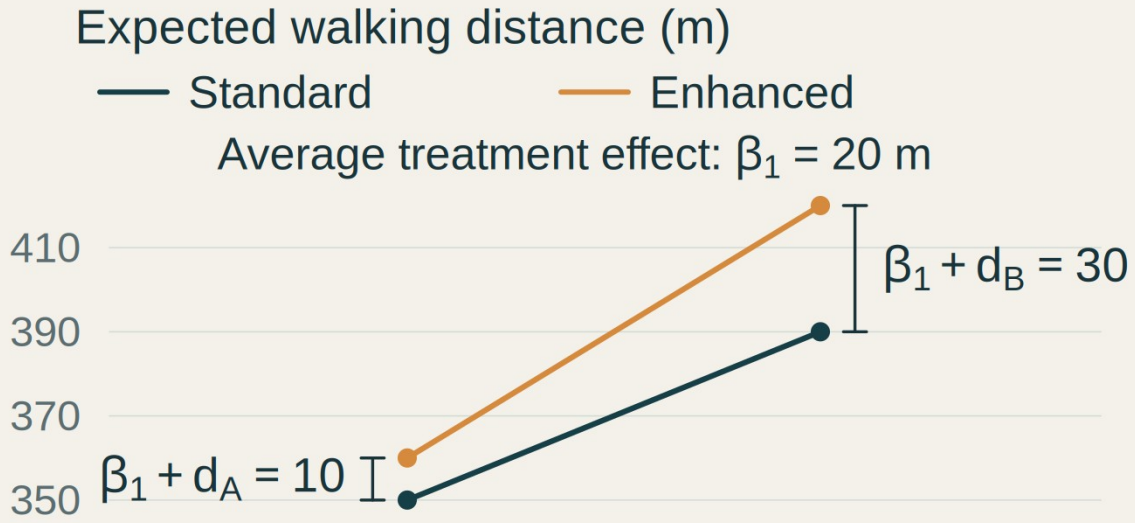
T_{ij} = 0 for standard care and 1 for enhanced care. i = patient; j = center.

Individual patient outcomes vary around their group mean through ε_{ij}.

Different treatment effects across centers

Hypothetical example: assume the four true group means are known.

Center	Standard	Enhanced	Row average
A	$\mu_{0A} = 350$	$\mu_{1A} = 360$	**355**
B	$\mu_{0B} = 390$	$\mu_{1B} = 420$	**405**
Column average	$\beta_0 = 370$	390	**380**



μ_{tj}: expected outcome for treatment t in center j

$$Y_{ij} = \underbrace{\beta_0 + c_j}_{\text{Standard-care mean in center } j} + \underbrace{(\beta_1 + d_j)T_{ij}}_{\text{Treatment contribution}} + \underbrace{\varepsilon_{ij}}_{\text{Individual deviation}}$$

T_{ij} = 0 for standard care and 1 for enhanced care. i = patient; j = center.

Individual patient outcomes vary around their group mean through ε_{ij}.

How should we represent differences between centers?

Beyond two centers, we still allow differences in baseline outcome and treatment effect.

FIXED CENTER EFFECTS

Give each included center its own baseline deviation and treatment-effect deviation.

Express the average treatment effect across the included centers.

RANDOM CENTER EFFECTS

Describe baseline deviations and treatment-effect deviations through a population distribution.

Express the average treatment effect in the population represented by those centers.

We start with fixed center effects and ask what we can learn about treatment in the included centers.

From known means to patient data

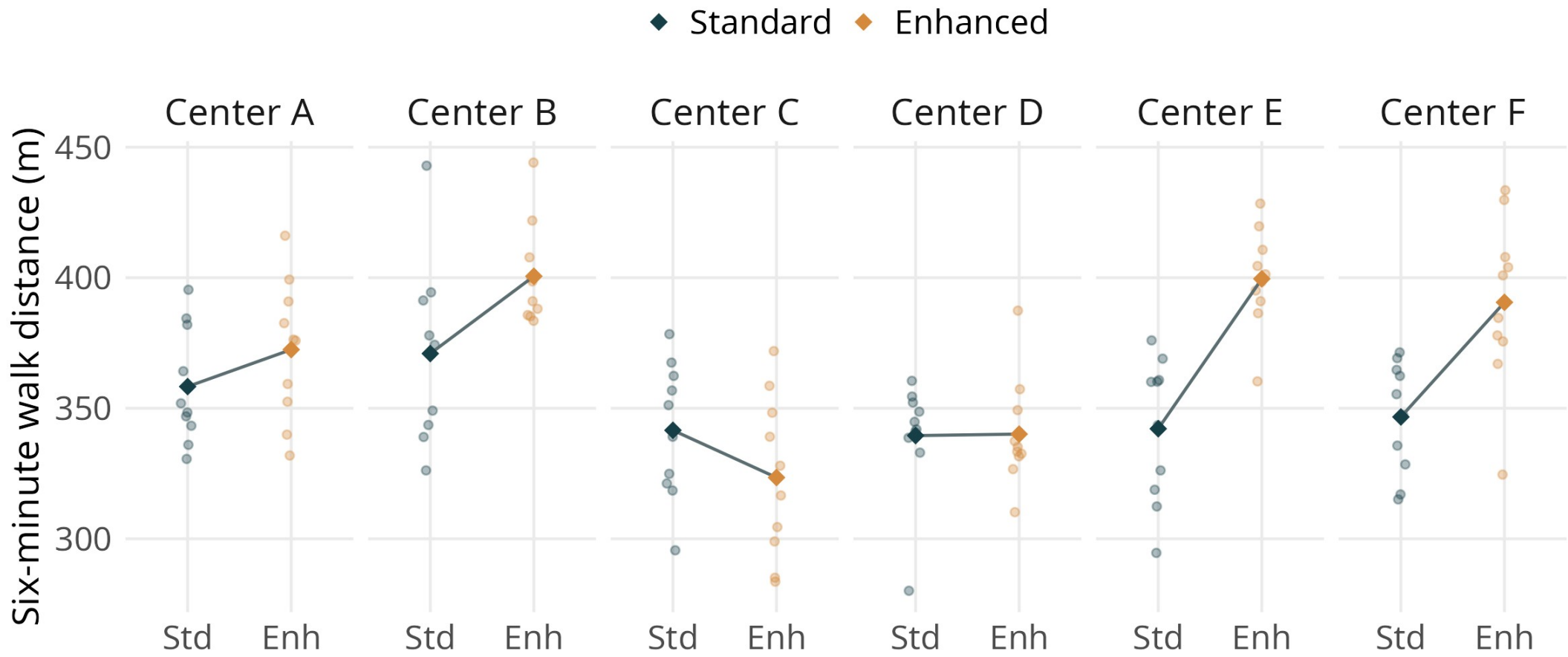
THE REHABILITATION TRIAL

- New simulated trial: 120 patients after knee replacement at six centers.
- Within each center: 10 patients randomized to standard care and 10 to enhanced care.
- Outcome: six-minute walk distance at 12 weeks.
- Fit the fixed-effects interaction model to estimate treatment effects in these centers.

The group means are now unknown and must be estimated from patient outcomes.



Explore outcomes by treatment and center



Each dot is a patient outcome; diamonds show sample means.
What differences do you notice in the treatment gaps across centers?

A fixed-effects model for treatment and center

$$Y_{ij} = \beta_0 + c_j + (\beta_1 + d_j)T_{ij} + \varepsilon_{ij}$$

T_{ij} = 0 for standard care and 1 for enhanced care. Center uses effects coding.

BASELINE OUTCOMES

β_0: average standard-care outcome across the included centers.

c_j: center j's deviation from that baseline average.

$$\sum_j c_j = 0$$

TREATMENT EFFECTS

β_1: average treatment effect across the included centers.

d_j: center j's deviation from the average treatment effect.

$$\sum_j d_j = 0$$

Each center receives equal weight in the averages. Center deviations are fixed, unknown parameters.

Fixed-effects interaction model: results

Term	Estimate (m)	Estimated SE (m)	95% CI (m)	p-value
Intercept (β_0)	349.9	3.31	343.3 to 356.4	< .001
Enhanced care (β_1)	21.3	4.68	12.0 to 30.6	< .001
Center A	8.4	7.41	-6.2 to 23.1	0.257
Center B	21.1	7.41	6.4 to 35.8	0.005
Center C	-8.3	7.41	-23.0 to 6.4	0.265
Center D	-10.3	7.41	-25.0 to 4.3	0.166
Center E	-7.7	7.41	-22.4 to 7.0	0.301
Enhanced × Center A	-7.1	10.48	-27.9 to 13.7	0.499
Enhanced × Center B	8.4	10.48	-12.4 to 29.1	0.427
Enhanced × Center C	-39.4	10.48	-60.1 to -18.6	< .001
Enhanced × Center D	-20.7	10.48	-41.4 to 0.1	0.051
Enhanced × Center E	36.2	10.48	15.4 to 56.9	< .001

Effects coding: Center F follows from the sum-to-zero constraints.

Testing interaction by comparing two models

Fit the full model → fit the reduced model → compare their fit.

Model	Residual sum of squares (RSS)	Residual df
Reduced: common treatment effect	90,694	113
Full: center-specific treatment effects	71,112	108

RSS: residual sum of squares. df: residual degrees of freedom.

$$F = \frac{(\mathrm{RSS}_{\mathrm{reduced}} - \mathrm{RSS}_{\mathrm{full}})/(\mathrm{df}_{\mathrm{reduced}} - \mathrm{df}_{\mathrm{full}})}{\mathrm{RSS}_{\mathrm{full}}/\mathrm{df}_{\mathrm{full}}} = \frac{(90{,}694 - 71{,}112)/(113 - 108)}{71{,}112/108} \approx 5.95$$

Numerator: improvement in fit per additional parameter.

Denominator: estimated individual-error variance from the full model.

F(5, 108) = 5.95, p < .001: strong evidence that treatment effects differ across centers.

Which treatment effect do we want to estimate?

SIMPLE TREATMENT EFFECTS

How do treatments compare within a particular center?

Compare the enhanced and standard means within each center.

$$\text{Effect in center } j: \quad \beta_1 + d_j$$

One treatment effect per center.

MARGINAL TREATMENT EFFECT

How do treatments compare on average across centers?

Average each treatment's mean equally across centers, then compare the averages.

$$\text{Average effect:} \quad \beta_1$$

With effects coding, the treatment deviations average to zero.

Estimated marginal means (EMMs) are model-based estimates of group means, averaged over other factors where appropriate. Comparing these means gives the treatment effects.

Estimating simple treatment effects with estimated marginal means

CODING AND EFFECTS

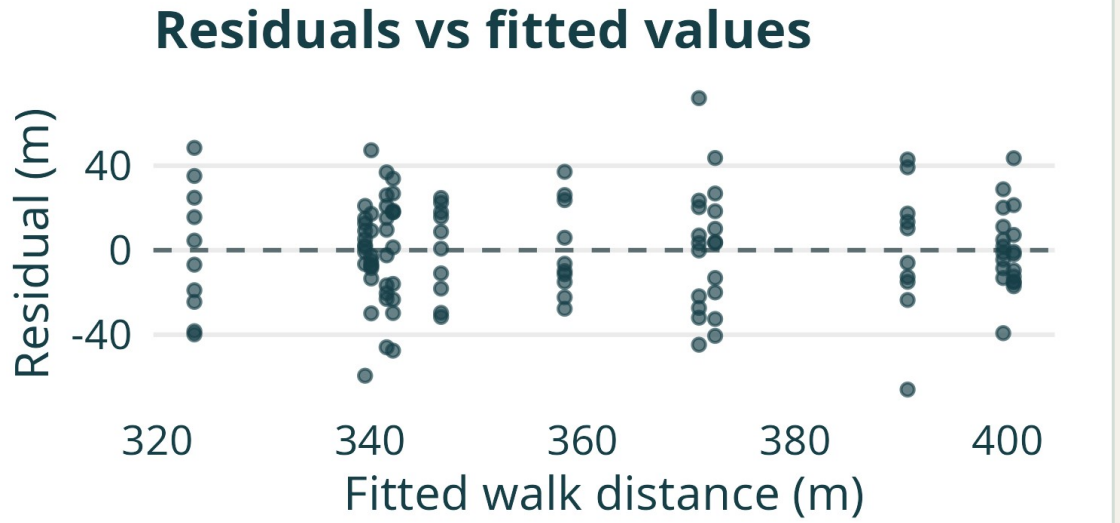
- Coding determines which effects appear directly as coefficients.
- Here, a center's effect combines the average effect with its treatment deviation.
- Predict both treatment means within each center, then compare them.

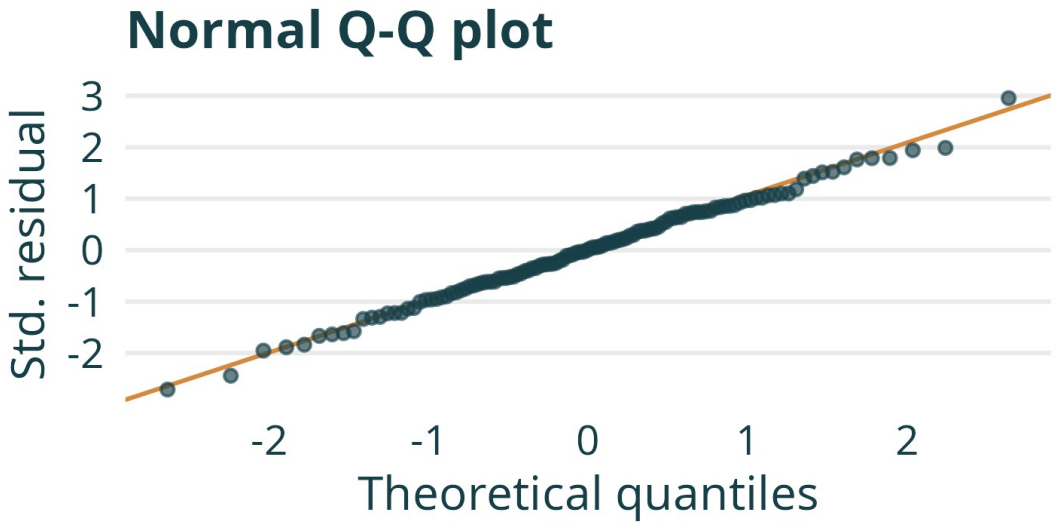
	Predicted means		Simple treatment effect		
Center	Standard mean	Enhanced mean	Difference	95% CI	p-value (Bonf.)
A	358.3	372.5	14.2	-8.6 to 36.9	1.000
B	370.9	400.6	29.6	6.9 to 52.4	0.067
C	341.6	323.5	-18.1	-40.8 to 4.7	0.707
D	339.5	340.1	0.6	-22.2 to 23.3	1.000
E	342.2	399.6	57.4	34.7 to 80.2	< .001
F	346.7	390.6	43.9	21.2 to 66.6	0.001

Means, differences and CIs in meters. Difference: enhanced minus standard. Pointwise 95% CIs; p-values adjusted across six comparisons (Bonferroni).

Equivalent coding schemes give different coefficients, but the same fitted means and treatment comparisons.

Model diagnostics: fixed-effects interaction





WHAT WE USE THIS PLOT FOR

Assess whether residual spread is similar across fitted walk distances.

WHAT WE USE THIS PLOT FOR

Assess whether the residual distribution is approximately Normal.

The plots show no clear change in residual spread and only small departures from the Normal reference line.

What if we leave out the interaction?

Estimated average treatment effect across the six centers

Model	Estimate (m)	Estimated SE (m)	95% CI (m)	p-value
Additive	21.3	5.17	11.0 to 31.5	< .001
With interaction	21.3	4.68	12.0 to 30.6	< .001

Same estimate: equal numbers in every treatment–center group make both models estimate the average of the six within-center treatment differences.

Larger estimated SE: without interaction, differences in treatment effects remain in the residuals, increasing the estimated residual variation.

In this balanced study, omitting interaction leaves the average treatment estimate unchanged but increases its estimated uncertainty.

What if center sizes differ?

WHAT THE MODELS ESTIMATE

- With our effects coding, the interaction model estimates the equally weighted average treatment effect, regardless of center sizes.
- If centers have equal sizes, the additive model estimates this same average.
- If center sizes differ, the additive model gives larger centers more weight.

ILLUSTRATIVE EXAMPLE

Center	Patients	Treatment effect (m)
Small	20	10
Large	80	30

Interaction model: 20 m

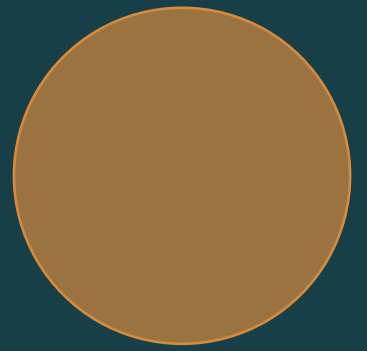
Additive model: 26 m

When treatment effects differ across centers, omitting interaction can change both the interpretation of the treatment coefficient and its estimated uncertainty.

PART 2

From included centers to a population





What changes if the study uses different centers?

THESE SIX CENTERS

Repeat the study with new patients in the same six centers.

Each center’s true treatment effect stays the same.

Uncertainty comes from estimating those six effects using patient outcomes.

A POPULATION OF CENTERS

Repeat the study with new centers sampled from a wider population.

The selected centers can have different true treatment effects.

Uncertainty also comes from which centers enter the study.

A population-average treatment effect requires accounting for variation in treatment effects between sampled centers.

Random treatment effects add a source of uncertainty

Let β_1 be the population-average treatment effect, with independent random center deviations:

$$d_j \overset{\text{iid}}{\sim} N(0, \sigma_d^2)$$

The deviations have expectation zero; their realized average need not be zero.

$$\widehat{\beta}_1 = \beta_1 + \frac{1}{J}\sum_{j=1}^{J} d_j + \frac{1}{J}\sum_{j=1}^{J} \eta_j$$

η_j is patient sampling noise, independent of the random center effects.

$$\mathrm{Var}(\widehat{\beta}_1) = \frac{\sigma_d^2}{J} + \frac{2\sigma^2}{Jn}$$

J = centers; n = patients per treatment per center; σ^2 = individual-error variance.

More patients and more centers answer different needs

AVERAGE IN THE INCLUDED CENTERS

Increasing patients per treatment makes each center's effect more precise.

$$\frac{2\sigma^2}{Jn} \longrightarrow 0 \quad \text{as } n \longrightarrow \infty$$

Even heterogeneous treatment effects can have a precisely estimated average within these centers.

AVERAGE IN THE POPULATION

More patients reduce individual noise, but the study still samples only J centers.

$$\frac{\sigma_d^2}{J} + \frac{2\sigma^2}{Jn} \longrightarrow \frac{\sigma_d^2}{J}$$

More centers reduce both variance terms. Six precisely measured centers still provide only six center effects.

These are sampling-variance formulas under the balanced model; fitted uncertainty also involves estimating the variance components.

The target determines which uncertainty belongs

FIXED INTERACTION

Target: the average effect in the included centers.

Patient sampling determines uncertainty around that fixed average.

Omitting the interaction can inflate the estimated uncertainty by leaving heterogeneity in the residuals.

RANDOM TREATMENT EFFECT

Target: the average effect in a population of centers.

Center sampling adds uncertainty about that population average.

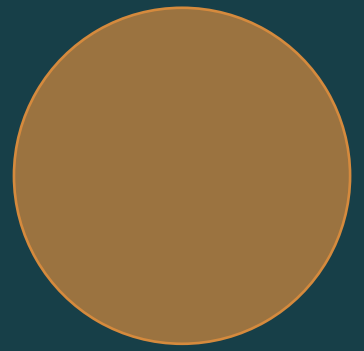
A random-intercept-only model can understate uncertainty because it omits treatment-effect variation.

Next: model a population of center-specific intercepts and treatment effects using random intercepts and slopes.

BACKUP

Further discussion





Why are the estimated standard errors almost identical?

MOST PRECISION COMES FROM WITHIN CENTERS

Both models compare enhanced and standard care within each center.

These comparisons are informative because the treatment groups share the same center baseline. Baseline differences between centers do not obscure these comparisons.
Both models obtain most of their precision from this same source.

BETWEEN CENTERS: LITTLE EXTRA PRECISION HERE

The random-intercept model can also use average outcomes from centers with different treatment proportions.

Centers differ substantially in baseline outcomes. A higher center average could reflect its baseline as well as its treatment allocation.

Under the random-intercept assumptions, this noisy evidence adds little precision: the estimated standard error barely decreases.

The additional evidence shifts the estimate slightly, but adds little precision: the estimated standard errors are 4.56 and 4.55 m.